

ELECTRIC POTENTIAL ENERGY

If 2 charges q_1 and q_2 , initially separated by a distance r , are moved apart, then work must be done. If both the charges have the same sign then the work will be negative. If they are of opposite signs the work will be positive. The energy represented by this work is thought of as being stored by the system of charges as electric potential energy.

$$W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} = U \quad = \text{electric potential energy}$$

For a dipole:

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1(-q_2)}{r_{12}}$$